

Java Streams

1. Filter Even Numbers

Problem:

Given a list of integers, use Stream API to filter out and collect all even numbers.

```
List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5, 6);  
// Output: [2, 4, 6]
```

2. Find the First String Starting with 'A'

Problem:

From a list of strings, find the first string that starts with the letter 'A'.

```
List<String> names = Arrays.asList("John", "Alice", "Bob", "Alex");  
// Output: "Alice"
```

3. Convert Strings to Uppercase

Problem:

Given a list of lowercase strings, convert them to uppercase using Stream API.

```
List<String> words = Arrays.asList("apple", "banana", "cherry");  
// Output: ["APPLE", "BANANA", "CHERRY"]
```

4. Count Empty Strings

Problem:

Count the number of empty strings in a list.

```
List<String> list = Arrays.asList("abc", "", "def", "", "ghi");  
// Output: 2
```

5. Find Max Value

Problem:

Use Stream API to find the maximum number in a list of integers.

```
List<Integer> nums = Arrays.asList(23, 5, 89, 16, 45);  
// Output: 89
```

6. Sort a List of Strings Alphabetically

Problem:

Sort a list of strings in alphabetical order using Stream API.

```
List<String> fruits = Arrays.asList("Banana", "Apple", "Mango", "Grapes");  
// Output: ["Apple", "Banana", "Grapes", "Mango"]
```

7. Group Words by Their Length

Problem:

Group a list of strings by their length.

```
List<String> words = Arrays.asList("one", "two", "three", "four", "six");  
// Output: {3=[one, two, six], 5=[three], 4=[four]}
```

8. Remove Duplicates

Problem:

Remove duplicate elements from a list of integers using Stream API.

```
List<Integer> numbers = Arrays.asList(1, 2, 2, 3, 4, 4, 5);  
// Output: [1, 2, 3, 4, 5]
```

9. Sum of All Odd Numbers

Problem:

Calculate the sum of all odd numbers from a list.

```
List<Integer> nums = Arrays.asList(1, 2, 3, 4, 5);  
// Output: 9 (1 + 3 + 5)
```

10. Find the Frequency of Each Word

Problem:

Given a list of strings, use Stream API to count the frequency of each word.

```
List<String> words = Arrays.asList("apple", "banana", "apple", "orange", "banana");  
// Output: {apple=2, banana=2, orange=1}
```

11. Find the Product of All Even Numbers Greater Than 10

Problem:

Given a list of integers, filter the numbers that are even and greater than 10, then find their product.

```
List<Integer> nums = Arrays.asList(5, 12, 18, 7, 22, 3);  
// Steps: filter (even && >10) → map (identity) → reduce (product)  
// Output: 12 * 18 * 22 = 4752
```

12. Total Length of Unique Words Starting With 'S'

Problem:

Given a list of strings, remove duplicates, filter words starting with 'S', map to their lengths, and reduce to get the total length.

```
List<String> words = Arrays.asList("Sun", "Sky", "Sea", "Sand", "Sky", "Sun", "Stone");  
// Output: 3 + 3 + 3 + 4 + 5 = 18
```

13. Concatenate Capitalized Names Longer Than 3 Characters

Problem:

Given a list of names, filter those longer than 3 characters, capitalize the first letter of each, and reduce to a single comma-separated string.

```
List<String> names = Arrays.asList("john", "al", "mark", "susan", "zoe");  
// Output: "John,Mark,Susan"
```

14. Sum of Squares of Odd Numbers from Nested Lists

Problem:

Given a list of lists of integers, flatten them, filter odd numbers, square them, and find the total sum.

```
List<List<Integer>> data = Arrays.asList(  
    Arrays.asList(1, 2, 3),  
    Arrays.asList(4, 5),  
    Arrays.asList(6, 7)  
);  
// Output: 1^2 + 3^2 + 5^2 + 7^2 = 1 + 9 + 25 + 49 = 84
```

15. Longest Word Starting With Vowel

Problem:

From a list of strings, filter words starting with a vowel, map to their lengths, and reduce to find the **length** of the longest such word.

```
List<String> words = Arrays.asList("apple", "banana", "orange", "umbrella", "ice", "grape");  
// Output: 8 ("umbrella")
```